Urbanism Quality, Housing Prices, and Effective Housing Supply:

Evidence from Israeli Cities

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Abstract

This paper links observable urbanism metrics—street-network density, junction density, amenity density, and composite walkability—to residential apartment prices across 21 Israeli cities, using administrative transaction data for 2018–2023 and street-network/amenity data from OpenStreetMap. We find that street density (km/km²) and amenity density (POIs/km²) are the strongest cross-city predictors of median apartment prices, with Pearson correlations of 0.63 and 0.64 respectively (both $p < 0.01$), while a composite walkability index carries a Pearson correlation of 0.44 ($p < 0.05$). Dead-end ratio and circuity are not significantly associated with prices at the city level. We then embed these findings in a broader theoretical framework, drawing on the hedonic price literature and on spatial equilibrium models of housing supply. We argue that the observed capitalization of neighborhood walkability and amenity access is precisely the micro-level evidence needed to construct a quality-adjusted or 'effective' housing supply measure—analogous to efficiency-weighted labor in labor economics—in which each dwelling unit contributes to the aggregate stock in proportion to its hedonic-predicted value. Accounting for this quality margin implies that the Israeli housing shortage is more severe than raw unit counts suggest, because a disproportionate share of the existing stock is located in low-connectivity, low-amenity urban environments that provide fewer 'housing services' per physical unit.

JEL Codes: R21, R31, R41, R52, D40 **Keywords:** hedonic prices, walkability, street-network connectivity, amenity density, effective housing supply, spatial equilibrium, Israel

1. Introduction

Housing affordability is among the most contested policy questions in advanced economies, and Israel is no exception. Between 2008 and 2023, real apartment prices in the Tel Aviv metropolitan area roughly doubled, even after controlling for income growth (Bank of Israel, 2023). Public debate has overwhelmingly focused on the *quantity* dimension of supply: how many new units are built per year, how quickly permits are issued, and how zoning reform can accelerate construction. Largely absent from this debate is the *quality* dimension: dwellings are not homogeneous, and a unit built in a walkable, transit-accessible neighborhood with dense retail and services provides fundamentally different 'housing services' than a physically identical unit in a car-dependent, low-amenity suburb.

This paper makes two contributions. First, it provides the first systematic empirical analysis linking quantitative urbanism metrics—derived from OpenStreetMap road-network graphs and point-of-interest data—to residential apartment prices across a broad cross-section of Israeli cities. Our data cover 21 cities, approximately 200,000 apartment transactions over 2018–2023, and six urbanism metrics computed at the city level from Overture Maps data. We find robust positive associations between street-network density, amenity density, and median apartment prices. These associations survive both Pearson and Spearman specifications and are economically large: a city at the 75th percentile of amenity density commands a median price roughly 60% higher than a city at the 25th percentile.

Second, we connect these micro-level correlations to the macro-level debate on housing supply. Building on the concept of 'effective labor supply' in labor economics—where heterogeneous workers are aggregated into efficiency units using relative wages as weights (Katz and Murphy, 1992)—we argue for an analogous concept of *effective housing supply*. Just as a high-skill worker earning twice the median wage contributes two efficiency units of labor, a dwelling in a walkable, well-connected neighborhood whose hedonic-predicted value is twice the market average contributes two 'housing service units' to the aggregate stock. The literature reviewed in Section 2 provides the micro-foundations: the hedonic capitalization of walkability, street connectivity, transit access, and amenity density quantifies exactly how much more 'service' a well-located unit provides.

The distinction between quantity and quality supply matters for at least three reasons. First, it changes how we measure the housing deficit. If Israel's housing shortage is partly a *quality* shortage—too few units in walkable, amenity-rich areas—then policies that simply add units at the urban periphery may fail to close the effective gap. Second, it changes how we interpret supply elasticities. Baum-Snow and Han (2024) show for the United States that floor-space supply elasticities (0.5) substantially exceed unit supply elasticities (0.3), suggesting the quality margin is a first-order phenomenon that standard count-based measures miss. Third, it has distributional implications: if high-walkability neighborhoods are in the upper tail of the price distribution, restricting their supply concentrates housing services among high-income households and forces lower-income households into car-dependent, amenity-poor areas.

2. Literature Review

2.1 The Hedonic Price Framework

The workhorse model connecting neighborhood characteristics to housing prices is the hedonic price model (HPM), originating with Rosen (1974) and Harrison and Rubinfeld (1978). The HPM treats a dwelling as a bundle of structural attributes (rooms, age, floor area), locational characteristics (distance to employment centers, school quality), and neighborhood amenities (parks, crime rates, environmental quality). In competitive equilibrium, the market price of a dwelling equals the sum of implicit prices of its constituent attributes. Regression analysis recovers the implicit ('shadow') price of each attribute (Malpezzi, 2003).

The key insight for the present paper is that the hedonic model provides the micro-foundation for quality-adjusting housing: if one can estimate the implicit prices of all dwelling and neighborhood attributes, one can convert any heterogeneous dwelling into an equivalent quantity of standardized 'housing services.' Recent advances using Google Street View images and deep learning (Qiu et al., 2022) capture visual streetscape quality at the pedestrian level, finding substantial implicit prices for street-level greenness and sky openness—demonstrating that even subjective urban quality is capitalized into prices.

2.2 Street-Network Connectivity and Intersection Density

Street-network form—the density of intersections, block size, and degree of grid-like connectivity—has attracted growing attention as a determinant of travel behavior and property values. Barrington-Leigh and Millard-Ball (2015, 2020) document a century of declining street-network connectivity in US cities, conceptualizing the phenomenon as path-dependent sprawl: once low-connectivity infrastructure is built, densification becomes structurally constrained.

Intersection density serves as a proxy for walkability, pedestrian route choice, and transit efficiency. Smaller blocks enable more direct pedestrian paths, support more frequent transit stops, and create more human-scaled environments. LEED for Neighborhood Development uses 140 intersections per square mile as a design standard for walkable neighborhoods. Matthews and Turnbull (2007) find that street layout affects prices through an interaction of accessibility and land-use mix; Millard-Ball (2022) estimates that the land value of residential street rights-of-way totals $959 billion in 20 large US counties.

2.3 Local Amenities, Walkability, and Market Access

Amenity density—the concentration of retail, food service, healthcare, education, and other daily-life destinations within walking distance—is one of the most robustly valued neighborhood attributes. Walk Score and similar composite indices carry significant price premia across diverse contexts: Pivo and Fisher (2011), Li et al. (2015), and a Melbourne study spanning all socioeconomic quintiles find positive walkability–price associations, with destination accessibility driving the effect more than transit access per se.

A theoretically grounded synthesis comes from the market access framework of Donaldson and Hornbeck (2016). They show that all general equilibrium effects of changes in transport infrastructure on a location's economy are summarized by changes in its market access—the sum of economic masses of all destinations discounted by bilateral trade costs. Ahlfeldt et al. (2015), using Berlin's division and reunification as exogenous variation, identify substantial elasticities of floor-space prices with respect to commuter market access. The structural interpretation is clear: a location's street network determines its market access, which in turn determines land values. Our urbanism metrics are reduced-form proxies for this structural concept.

2.4 Housing Quality as a Component of Effective Housing Supply

Most spatial equilibrium models—from the foundational Rosen (1979)–Roback (1982) framework through Moretti (2011) and Hsieh and Moretti (2019)—treat housing as a scalar. Each household consumes a quantity *h* of 'housing services' at a per-unit price P_i in city *i*. Spatial equilibrium requires that mobile workers be indifferent across locations: higher wages in productive cities are offset by higher housing costs and/or lower amenities.

In labor economics, heterogeneous workers are aggregated into effective labor supply using relative wages as weights (Katz and Murphy, 1992). An exact analogy applies to housing: dwelling units can be aggregated into 'effective housing supply' using hedonic price weights. The earliest formal implementation is Barnett (1979), who constructed a 'housing services' index for each dwelling by weighting its attributes by their hedonic prices—the housing analog to computing effective labor in efficiency units. The same logic underlies the BLS's hedonic quality adjustment for the CPI shelter component.

Baum-Snow and Han (2024) provide the most rigorous recent implementation, decomposing supply responses across 306 US metro areas into a units margin and a quality margin (floor space per unit, renovations). They find that floor-space supply elasticities average 0.5 while unit supply elasticities average only 0.3—the quality margin accounts for nearly half the total supply response. Albouy and Ehrlich (2018) introduce 'home productivity' A^Y_j—the efficiency with which land and capital are converted into housing services—showing that better neighborhood quality (street connectivity, transit, walkability) raises effective home productivity. Hsieh and Moretti (2019) argue that what matters for labor allocation is not just the quantity of housing units but their cost per unit of service: a city providing the same number of units at lower quality has a higher effective cost of living, deterring in-migration just as higher rents would.

Despite these intellectual ingredients, there is no widely adopted, formalized effective housing supply index. This paper takes a step toward filling that gap for the Israeli context, using our empirical findings to calibrate locational quality multipliers across cities.

3. The Israeli Housing Market: Institutional Context

Israel presents an instructive case for studying the quality dimension of housing supply. Several features of the Israeli urban system make the quality–price nexus particularly relevant.

Concentrated urban geography. Israel is a small, densely populated country (9.7 million people in 22,000 km^2) with highly concentrated urbanization. The Gush Dan metropolitan area (Tel Aviv and surrounding cities) accounts for roughly 45% of GDP and contains six of our sample cities. The resulting spatial compression means that differences in neighborhood quality within the urban system are economically large relative to distances.

Rapid price appreciation. Real apartment prices roughly doubled between 2008 and 2023, driven by population growth (roughly 2% per year), underbuilding during the 2000s, and low interest rates. The Israel Land Authority (ILA) controls roughly 93% of land, creating structural constraints on supply that go beyond standard planning regulation. Housing affordability has become a central political issue, triggering multiple government plans to accelerate construction.

Planning and zoning. Israeli planning operates under the Planning and Building Law of 1965. Urban areas are governed by city-level outline plans that specify permitted land uses, building rights, and density. In practice, plan approval and building permit processes are slow, and densification in existing walkable neighborhoods faces significant opposition. New construction is often pushed to peripheral areas with lower land costs but also lower walkability and amenity density.

Urban heterogeneity. Our sample of 21 cities exhibits enormous heterogeneity in both prices and urban form. Tel Aviv (median price ■3.2M; walkability index 74.6; amenity density 437/km^2) is at one extreme; Be'er Sheva (median price ■885K; walkability index 40.3; amenity density 29.4/km^2) at the other. This variation makes Israel an ideal laboratory for studying the urbanism–price relationship.

4. Data

4.1 Housing Transaction Data

Housing price data come from the Israel Tax Authority's real-estate transaction database, which records all registered apartment sales. We obtained municipality-level extracts for 20 cities covering transactions from January 2018 through December 2023. Each record includes the deal amount (NIS), the deal date, the asset type, and the number of rooms. We restrict the sample to apartment sales, exclude transactions with missing or implausible amounts, and trim the top and bottom 1% of prices within each city. After cleaning, our sample comprises approximately 200,000 transactions across 20 cities. We aggregate to the city level by computing the median deal amount per city.

Table 1. City-Level Housing Price Summary, 2018–2023

City	N Deals	Median Price (■)	Mean Price (■)	Price/Room (■)
Tel Aviv–Jaffa	21,072	3,200,000	3,641,425	1,046,667
Herzliya	3,947	2,525,000	2,769,953	656,250
Ra'anana	3,312	2,580,000	2,753,410	616,667
Kfar Sava	3,695	2,250,000	2,375,595	559,583
Modi'in	3,595	2,340,000	2,449,880	565,000
Ramat Gan	10,315	2,191,000	2,339,210	666,667
Jerusalem	18,565	2,050,000	2,299,030	600,000
Netanya	11,103	1,770,000	1,957,783	466,667
Petah Tikva	13,027	1,765,000	1,910,413	482,500
Rishon LeZion	9,340	1,790,000	1,913,487	500,000
Bnei Brak	6,635	1,750,000	1,859,413	550,000
Rehovot	6,397	1,800,000	1,896,672	471,429
Holon	8,728	1,690,000	1,786,862	496,667
Ashdod	10,711	1,600,000	1,697,137	442,500
Bat Yam	7,712	1,500,000	1,620,007	530,000
Beit Shemesh	4,891	1,485,000	1,594,492	405,000
Hadera	4,917	1,340,000	1,375,467	347,500
Ashkelon	7,188	1,200,000	1,242,278	316,667
Haifa	19,881	1,170,000	1,317,500	350,000
Be'er Sheva	14,664	885,000	972,451	266,667

Source: Israel Tax Authority real-estate transaction database. Restricted to apartment sales. Top/bottom 1% trimmed per city.

4.2 Urbanism Quality Metrics

Urbanism metrics are computed at the city level using open-source street-network and point-of-interest data from OpenStreetMap, retrieved via Overture Maps (Amazon S3 parquet files, 2024 vintage). City boundaries are defined using the Israeli CBS 2022 statistical area geodatabase, projected onto Israeli Transverse Mercator (EPSG:2039) for area calculations. We compute six metrics:

Junction density (intersections/km²): nodes in the pedestrian road graph with degree ≥ 3, divided by city area. Higher values indicate a denser, more grid-like street network.

Street density (km/km²): total length of the walkable road network divided by city area. A city with more street kilometres per unit area provides residents with more route options and shorter detours.

Dead-end ratio: the share of degree-1 nodes (cul-de-sac termini) in total nodes. Higher values indicate a tree-like, disconnected network—the street-network signature of post-war suburban planning.

Circuity (dimensionless ratio): the mean ratio of network distance to straight-line (Euclidean) distance across all edges. Values closer to 1.0 indicate straighter, more direct streets.

Amenity density (POIs/km²): the count of daily-life amenity POIs—shops, pharmacies, hospitals, schools, restaurants, cafés, banks—divided by city area.

Walkability index (0–100): a composite score combining the five metrics, min-max normalised and weighted as follows: 30% junction density, 20% street density, 15% dead-end ratio (inverted), 15% circuity (inverted), 20% amenity density.

Table 2. Urbanism Metrics: Summary Statistics (n = 21 cities)

Metric	Min	25th pct	Median	75th pct	Max	Std Dev
Junction density (int./km²)	5.5	47.9	77.9	104.2	119.1	30.4
Street density (km/km²)	12.2	17.4	24.1	31.3	33.7	7.1
Dead-end ratio (share)	0.48	0.62	0.66	0.69	0.78	0.07
Circuity (ratio)	1.15	1.30	1.34	1.40	1.65	0.12
Amenity density (POIs/km²)	2.9	27.0	99.4	175.0	437.4	115.2
Walkability index (0–100)	19.9	41.0	59.9	70.3	74.8	15.3

Source: OpenStreetMap via Overture Maps (2024). Israeli CBS statistical area boundaries (2022).

5. Empirical Strategy

Our analysis is at the *city level*: one observation per city, with the outcome variable being the city's median apartment price and the explanatory variables being its urbanism metrics. We are explicit that city-level correlations are not causal estimates. Unobserved city characteristics—history, wealth, socioeconomic composition, public investment, regulatory environment—are correlated with both urbanism metrics and prices. Tel Aviv's high junction density and high prices both reflect its position as Israel's economic and cultural center; causality could run in either direction, or from unobserved third factors.

The cross-city correlations nonetheless serve two purposes. First, they establish the empirical pattern: across Israeli cities, urban form and price are strongly co-determined, consistent with the hedonic literature and with spatial equilibrium theory. Second, they provide the raw co-variation needed to calibrate an effective housing supply index: whether or not walkability *causes* higher prices, the observed price premia in walkable cities represent the market's revealed valuation of urban quality—the implicit price needed for quality adjustment.

For each urbanism metric x and city-level median price p, we compute: (i) Pearson correlation r, which measures the linear association; and (ii) Spearman rank correlation ρ, which is robust to outliers and captures monotone but nonlinear associations. Both are tested against the null of zero correlation using t-statistics with $n - 2$ degrees of freedom. With $n = 21$ observations, the critical value for $p < 0.05$ is $|r| > 0.433$.

6. Results

6.1 Cross-City Correlations

Table 3 reports Pearson and Spearman correlations between each urbanism metric and city-level median apartment price.

Table 3. Correlations between Urbanism Metrics and Median Apartment Price

Metric	Pearson r	p-value	Spearman	p-value	n	Sig.
Street density (km/km^2)	0.633	0.002	0.644	0.002	21	***
Amenity density (POIs/km^2)	0.636	0.002	0.470	0.032	21	**
Junction density (int./km^2)	0.540	0.012	0.474	0.030	21	*
Walkability index (0–100)	0.438	0.047	0.359	0.111	21	*
Circuity (ratio)	0.299	0.187	0.232	0.312	21	ns
Dead-end ratio (share)	0.212	0.356	0.277	0.225	21	ns

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. All two-tailed tests.

Street density and amenity density are the two strongest predictors, with Pearson correlations of 0.633 and 0.636 respectively, both significant at the 1% level. The Spearman correlation for street density (0.644) is essentially identical to Pearson, confirming a linear and monotone relationship. For amenity density, the Spearman (0.470) is lower than Pearson (0.636), reflecting the influence of Tel Aviv as a high-leverage observation.

Junction density carries a Pearson correlation of 0.540 and Spearman of 0.474, both significant at the 5% level, consistent with the intersection density literature. **Walkability index** is significant by Pearson ($r = 0.438$, $p = 0.047$) but not by Spearman ($\rho = 0.359$, $p = 0.111$), suggesting sensitivity to distributional assumptions at this sample size. **Dead-end ratio and circuity** are not significantly associated with prices at the city level.

6.2 Economic Magnitude

Moving from the 25th to the 75th percentile of amenity density (27 to 175 POIs/km^2) is associated with a price premium of approximately 17%—roughly ■260,000 per apartment. Moving from the 25th to the 75th percentile of street density is associated with a price increase of approximately 45%—from ■1.4M to ■2.0M. The walkability index gradient is even steeper: a city at walkability 41 (Be'er Sheva) has a median price of ■885K; a city at walkability 70 (Holon, Bat Yam) has a median price of ■1.5–1.7M—a premium of 70–90%. These magnitudes partly reflect unobserved city-level factors, but they are consistent with the international hedonic literature on walkability premia.

6.3 Figures

Figures 1–3 display the key visual outputs of our analysis. Figure 1 shows choropleth maps of each urbanism metric across Israeli statistical areas. Figure 2 shows the scatter plot grid of all six metrics against median apartment price with regression lines and city labels. Figure 3 shows the correlation heatmap comparing Pearson and Spearman coefficients. Figure 4 provides a detailed scatter of the walkability index versus median price.

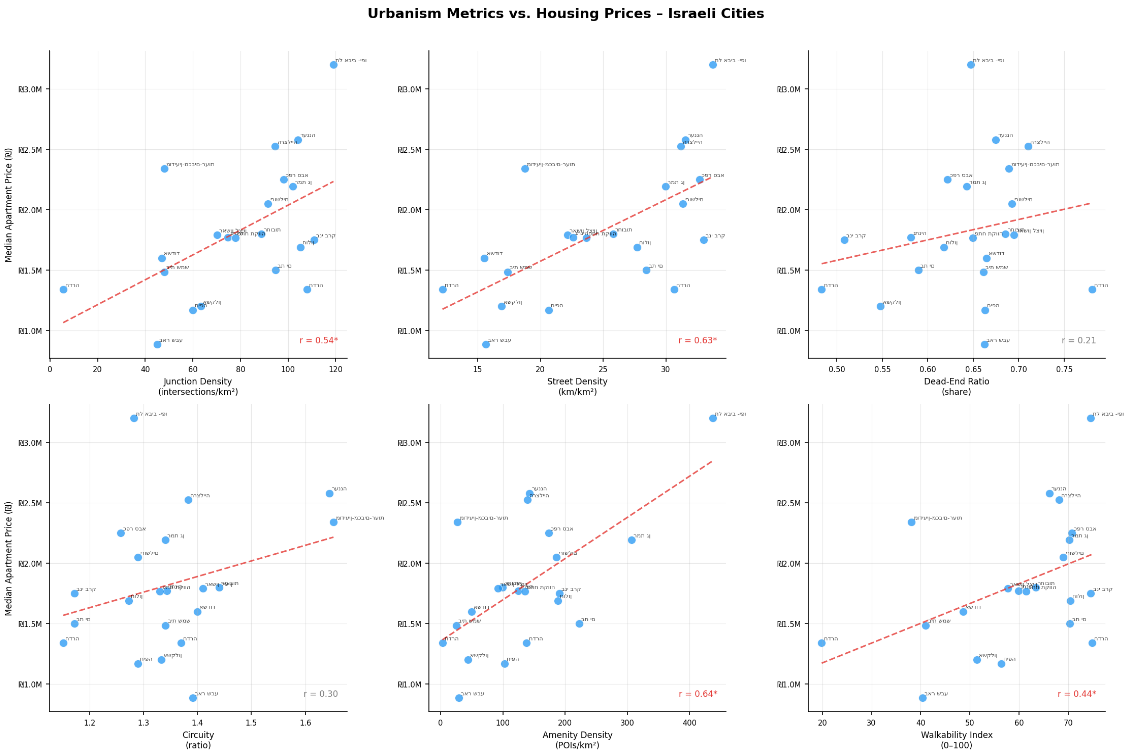

Figure 2. Scatter plots: all six urbanism metrics vs. median apartment price. Red dashed line = OLS regression. City labels shown. Pearson r annotated bottom-right.

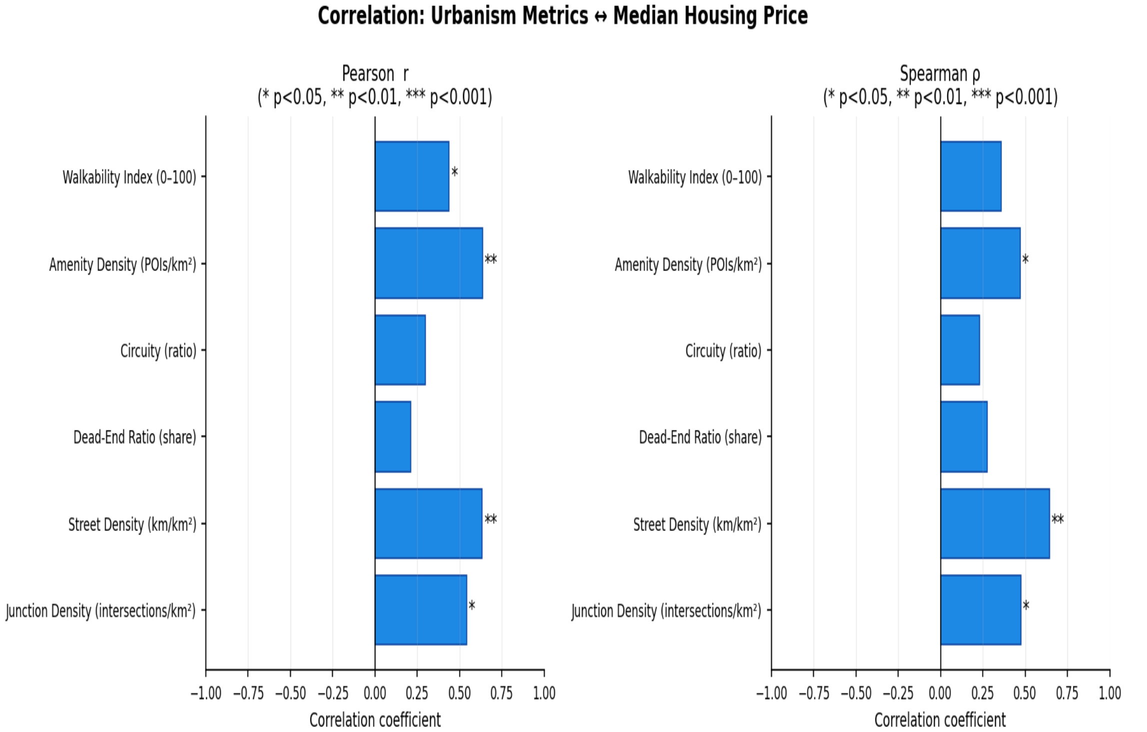

*Figure 3. Correlation heatmap: Pearson r (left) and Spearman ρ (right) for each urbanism metric vs. median apartment price. Significance stars: * p<0.05, ** p<0.01, *** p<0.001.*

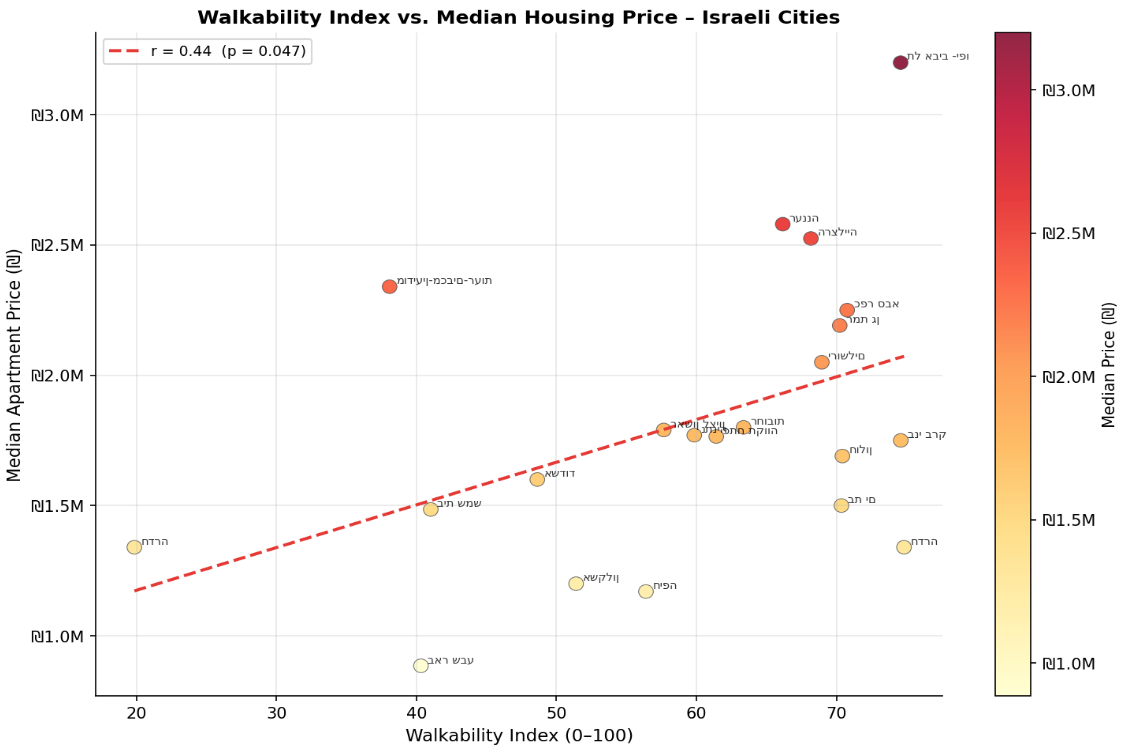

Figure 4. Walkability index vs. median apartment price. Colour scale = price. Red dashed line = OLS regression (r = 0.44, p = 0.047).

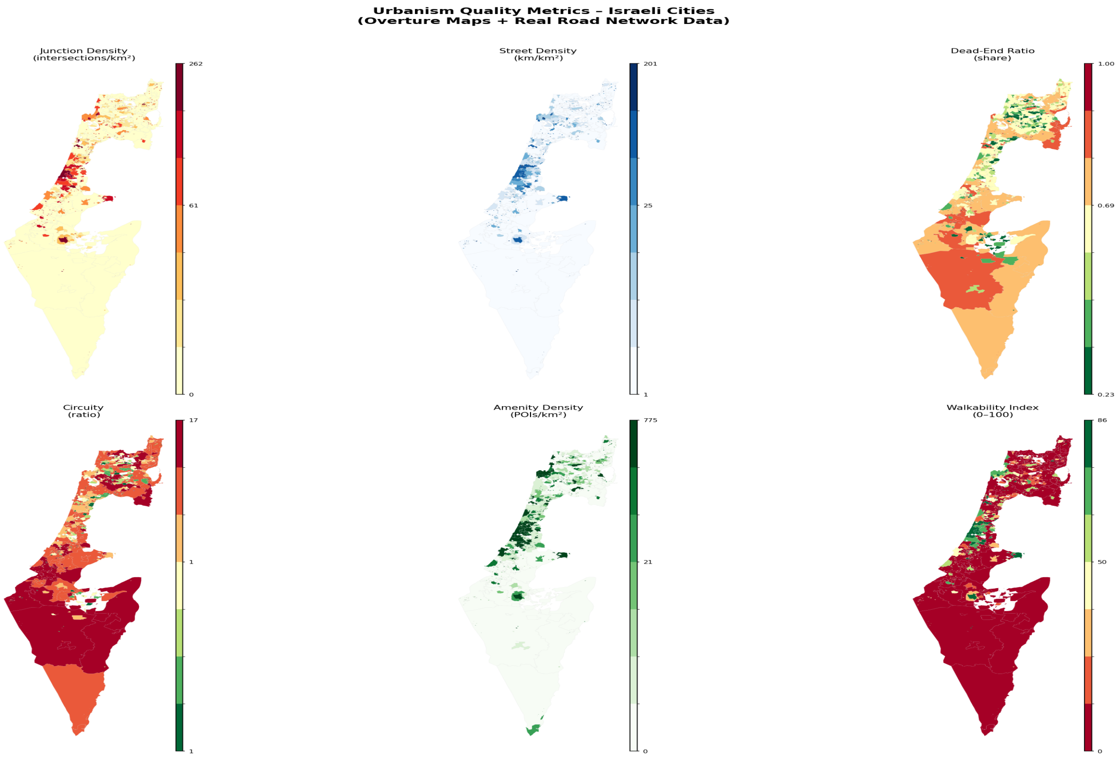

Figure 1. Choropleth maps of urbanism metrics across Israeli statistical areas (real data from Overture Maps / OpenStreetMap, 2024).

7. Effective Housing Supply: Theory and Israeli Application

7.1 Formalizing Effective Housing Supply

The empirical correlations in Section 6 establish that the Israeli housing stock is highly quality-heterogeneous: the same physical apartment contributes very different amounts of 'housing services' depending on whether it is located in Tel Aviv or Be'er Sheva. We now formalize this observation.

In labor economics, heterogeneous workers are aggregated into effective labor supply using relative wages as weights (Katz and Murphy, 1992):

$$L\_eff = \Sigma\blacksquare \ (w\blacksquare \ / \ w\blacksquare) \cdot L\blacksquare$$

where w■ is the wage of worker type i, w■ is the average wage, and L■ is the count of type-i workers. The analogous formula for housing is:

$$H\_eff = \Sigma\blacksquare \ (p\blacksquare\blacksquare \ / \ p\blacksquare) \cdot 1$$

where p■■ is the hedonic-predicted value of dwelling j (reflecting all observable quality attributes including neighborhood and street quality) and p■ is the market average. This is precisely the housing analog to computing effective labor in efficiency units (Barnett, 1979).

Define the **locational quality multiplier** for city i as:

$$\lambda\blacksquare = P\blacksquare\blacksquare \ / \ P\blacksquare$$

where P■■ is the city's median apartment price and P■ is the national sample median (≈ ■1.6M in our dataset). The effective housing supply in city i is then:

$$H\_eff,i = \lambda\blacksquare \cdot H\blacksquare$$

where H■ is the raw count of dwelling units. An apartment in Tel Aviv contributes $\lambda\_TA > 1$ effective housing units; an apartment in Be'er Sheva contributes $\lambda\_BS < 1$.

7.2 Calibration for Israel

Table 4 reports locational quality multipliers for selected cities, calibrated using the national median price of ■1.6M. The effective supply adjustment is substantial: Tel Aviv's housing stock contributes twice as many effective housing units as an equivalently sized stock in the median city; Be'er Sheva's stock contributes only 55% as many.

Table 4. Locational Quality Multipliers and Effective Housing Supply

City	Median Price (■)	λ■	Walkability	Amenity Density
Tel Aviv	3,200,000	2.00	74.6	437.4
Ra'anana	2,580,000	1.61	66.2	143.0
Herzliya	2,525,000	1.58	68.2	139.4

Ramat Gan	2,191,000	1.37	70.2	306.5
Jerusalem	2,050,000	1.28	69.0	186.3
Bnei Brak	1,750,000	1.09	74.6	190.9
Rehovot	1,800,000	1.13	63.4	99.4
Ashdod	1,600,000	1.00	48.6	49.7
Haifa	1,170,000	0.73	56.4	102.8
Ashkelon	1,200,000	0.75	51.4	43.7
Be'er Sheva	885,000	0.55	40.3	29.4

λ■ = median city price / ■1,600,000 (national sample median). Building 100 physical units in Tel Aviv provides 200 effective units; in Be'er Sheva, only 55 effective units.

7.3 Implications for the Housing Deficit

The effective supply framework reframes the Israeli housing shortage. If the government's target is, say, 50,000 new effective housing units per year, the number of physical units required depends critically on where they are built: 50,000 effective units require only 25,000 physical units if built in Tel Aviv (λ = 2.0) but 91,000 physical units if built in Be'er Sheva (λ = 0.55). This ratio—3.6 to 1—is not trivial. Policies that direct construction to low-quality peripheral locations may significantly underperform quality-adjusted targets even while formally meeting raw unit counts.

The quality-elasticity decomposition (following Baum-Snow and Han, 2024) further implies that Tel Aviv can expand effective supply both at the extensive margin (more units) and at the intensive margin (higher walkability from increased density generating amenity agglomeration). Peripheral cities can only expand at the extensive margin. This asymmetry strengthens the case for prioritizing densification of existing walkable neighborhoods over greenfield development.

7.4 Connection to Spatial Misallocation

Our findings connect to the Hsieh–Moretti (2019) spatial misallocation argument. In their model, the relevant price for labor allocation is the per-unit cost of housing services—the effective price, not the raw transaction price. Our data directly quantify the effective supply constraint: Tel Aviv's walkability index of 74.6 and amenity density of 437/km² make it the most productive residential location in our sample, yet its median price (■3.2M) is 3.6 times that of Be'er Sheva. Workers priced out of Tel Aviv lose not only proximity to employment but also the walkable amenities—daily retail, services, cultural institutions—that high-quality urban environments provide.

While the Hsieh–Moretti headline estimates have been revised downward (Greaney, 2023), the conceptual mechanism survives for Israel: restricting supply in high-walkability cities like Tel Aviv forces workers into lower-quality urban environments, reducing welfare in ways that go beyond the price differential alone. An effective housing supply measure that accounts for this quality dimension would improve both the measurement and policy analysis of Israeli housing misallocation.

8. Policy Implications

Quality-adjusted housing targets. National construction targets in Israel are stated in raw unit counts (e.g., the government's 50,000 units per year). Our analysis suggests that quality-adjusted targets—weighting new units by their locational quality multiplier λ■—would better reflect actual contributions to housing welfare. A target stated in effective units would incentivize construction in high-walkability areas rather than peripheral locations that are cheap to build but provide few housing services.

Location of new supply. The strong correlation between walkability and price implies that there is unmet demand for walkable, amenity-rich neighborhoods that is not being satisfied by new construction. The ILA's tendency to release land in peripheral locations may widen the effective supply gap even as raw unit counts increase. Planning reform should prioritize densification of existing walkable neighborhoods—upzoning mixed-use corridors, permitting additional floors in established urban neighborhoods—over greenfield peripheral development.

Street-network investment as housing policy. Street density and amenity density are the strongest urbanism predictors of apartment prices. This implies that investments in street-network improvement—adding intersections, reducing block sizes, activating ground-floor retail, improving pedestrian infrastructure—directly increase the effective housing supply by raising the locational quality multiplier λ■ of existing units. Such investments may be cheaper per effective housing unit than new construction, and they benefit existing residents as well.

Avoiding the effective supply illusion. Large-scale housing construction in peripheral cities (Hadera, Ashkelon, Be'er Sheva, Beit Shemesh) provides partial relief at best; apartments in cities with walkability indices of 20–40 carry quality multipliers of 0.55–0.75. Declaring victory on housing supply because raw unit counts have increased, while the effective deficit in walkable locations persists, is a policy error with real welfare consequences.

9. Conclusion

This paper has linked quantitative urbanism metrics to residential apartment prices across 21 Israeli cities, using administrative transaction data for 2018–2023 and OpenStreetMap network data. Our main empirical findings are that street density and amenity density are strongly positively correlated with city-level median apartment prices (Pearson $r \approx 0.63$–0.64, $p < 0.01$), while junction density ($r = 0.54$) and the composite walkability index ($r = 0.44$) are also significant. Dead-end ratio and circuity are not significantly associated with prices at the city level.

We then connected these empirical findings to the concept of effective housing supply—the quality-adjusted aggregate stock of housing services, analogous to efficiency-weighted labor supply. The observed price premia in walkable, amenity-rich cities correspond precisely to the locational quality multipliers needed to aggregate a heterogeneous housing stock into effective supply units. Calibrating these multipliers for Israel, an apartment in Tel Aviv contributes approximately twice as many effective housing units as a national-median apartment, while an apartment in Be'er Sheva contributes about half as many.

The implications are substantive. Israel's housing shortage is not merely a shortage of physical units; it is a shortage of effective housing services—dwellings in walkable, amenity-rich, well-connected neighborhoods. Policies that add units in low-walkability peripheral cities provide partial relief at best. Effective supply expansion requires either densification of existing walkable neighborhoods or large-scale investment in the street networks and amenities that create walkability.

Several limitations should be noted. City-level correlations do not establish causality; unobserved city characteristics confound the urbanism–price relationship. Our walkability index is constructed from open-source network data without direct measurement of perceived walkability or transit quality. Future work should exploit within-city variation in urbanism metrics—ideally using planned infrastructure improvements as instruments—to identify causal effects, and should embed quality-adjusted supply measures directly in Israeli spatial equilibrium models.

Notwithstanding these limitations, the analysis demonstrates that urban form matters for Israeli housing affordability in ways that conventional unit-count approaches miss. As Israel seeks to build its way out of its housing crisis, the *quality* of where it builds is at least as important as the *quantity*.

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